

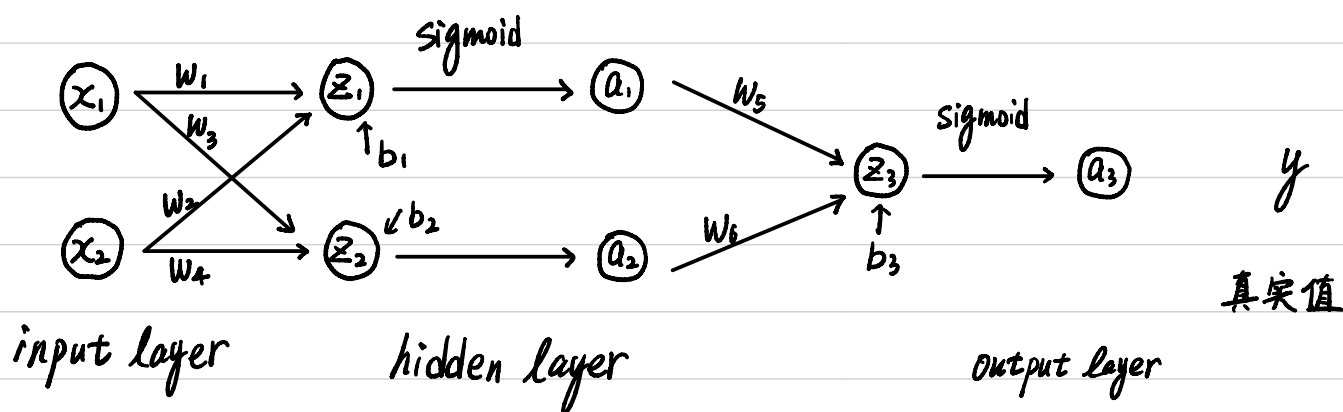
机器学习—台大李宏毅

反向传播

用一个实例来说明反向传播的原理

→ 来源 b 站“快速学会 BP 神经网络正向传播与反向传播的数学表达”

1. Build a neural network



2. Forward Pass

$$\textcircled{1} \begin{cases} z_1 = w_1 x_1 + w_2 x_2 + b_1 \\ z_2 = w_3 x_1 + w_4 x_2 + b_2 \end{cases} \quad \textcircled{2} \begin{cases} a_1 = \frac{1}{1 + e^{-z_1}} \\ a_2 = \frac{1}{1 + e^{-z_2}} \end{cases}$$

$$\textcircled{3} z_3 = w_5 a_1 + w_6 a_2 + b_3 \quad \textcircled{4} a_3 = \frac{1}{1 + e^{-z_3}}$$

3. Backward Pass

用 MSE
损失函数: $L = \frac{1}{2} (y - a_3)^2$

反向传播的目的是更新 weight 来减少损失函数的一个数值, 这里采用梯度下降法, 对参数进行更新

$$\frac{\partial L}{\partial w_6} = \frac{\partial L}{\partial a_3} \cdot \frac{\partial a_3}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_6} \quad \frac{\partial L}{\partial w_4} = \frac{\partial L}{\partial a_3} \cdot \frac{\partial a_3}{\partial z_3} \cdot \frac{\partial z_3}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_4}$$

$$\frac{\partial L}{\partial w_5} = \frac{\partial L}{\partial a_3} \cdot \frac{\partial a_3}{\partial z_3} \cdot \frac{\partial z_3}{\partial w_5} \quad \frac{\partial L}{\partial w_3}, \frac{\partial L}{\partial w_2}, \frac{\partial L}{\partial w_1} \text{ 以此类推}$$

$$\frac{\partial L}{\partial b_3} = \frac{\partial L}{\partial a_3} \cdot \frac{\partial a_3}{\partial z_3} \cdot \frac{\partial z_3}{\partial b_3} \quad \frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial a_3} \cdot \frac{\partial L}{\partial z_3} \cdot \frac{\partial z_3}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} \cdot \frac{\partial z_2}{\partial b_2}; \frac{\partial L}{\partial b_1} \text{ 同理}$$

第二步：参数更新

$$W_{\text{new}} = W_{\text{old}} - \eta \frac{\partial \text{Loss}}{\partial W}$$

在反向传播中计算出来的
梯度用于更新参数